

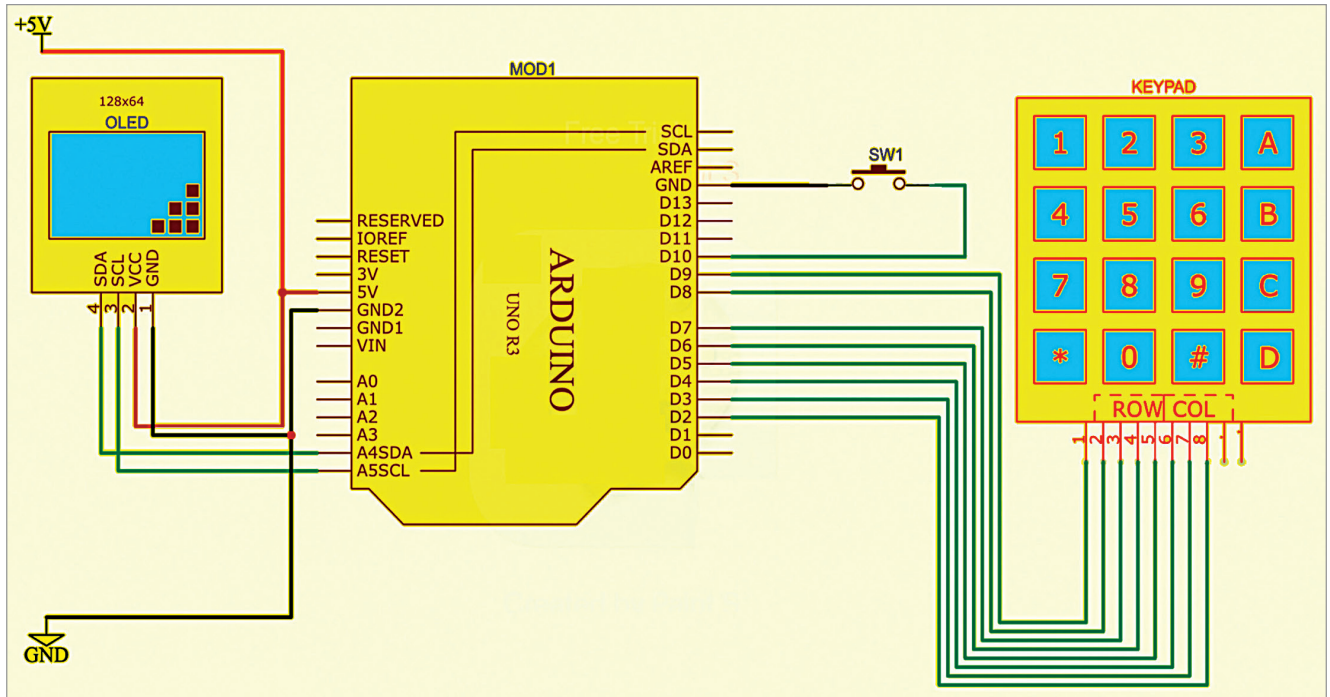


Fig. 2: Circuit diagram

```

sketch_apr23a.ino
1  //RAKESH JAIN PROGRAM FOR ARDUINO BASED DUAL MODE SCIENTIFIC CALCULATOR
2  #include <Wire.h>
3  #include <Adafruit_GFX.h>
4  #include <Adafruit_SSD1306.h>
5  #include <Keypad.h>
6  #include <math.h> // Include math functions
7  #define SCREEN_WIDTH 128
8  #define SCREEN_HEIGHT 64
9  #define BUTTON_PIN 10 // Push button for switching modes
10 Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);
11 // Keypad Configuration
12 const byte ROWS = 4;
13 const byte COLS = 4;
14 char keys[ROWS][COLS] = {
15   {'1', '2', '3', 'A'},
16   {'4', '5', '6', 'B'},
17   {'7', '8', '9', 'C'},
18   {'*', '0', '#', 'D'}
19 };
20 byte rowPins[ROWS] = {9, 8, 7, 6};
21 byte colPins[COLS] = {5, 4, 3, 2};
22 Keypad keypad = Keypad(makeKeymap(keys), rowPins, colPins, ROWS, COLS);
23 String input = "";
24 char lastOperator = '\0';
25 float num1 = 0, num2 = 0;
26 bool isNewNumber = true;
27 bool scientificMode = false; // Toggle between normal and scientific mode
28 void setup() {
29   Serial.begin(9600);
30   pinMode(BUTTON_PIN, INPUT_PULLUP); // Internal pull-up resistor
31   if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
32     Serial.println(F("SSD1306 initialization failed"));
33     for (;;);
34   }
35   displayWelcomeMessage();
36 }
37 void loop() {
38   // Check if the button is pressed

```

Fig. 3: Snippet of the source code

of the Arduino. When the button (SW1) is pressed, the calculator switches to scientific mode, displaying 'scientific mode' on the OLED screen.

The Arduino code is structured to detect keypad input, process numbers and operators, and update the OLED display accordingly. The loop() function continuously checks the state of the push button, toggling between standard and scientific modes. When a key is pressed, the program determines the appropriate operation based on the active mode and

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